

Bayesian Analysis

Bayes' Rule

Bayes' rule (or Bayes' theorem) was named after [Thomas Bayes](#) (1701-1761).

Letting A and B be two events,

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

- Math to English translation:
 - $P(A)$: probability that event A happens
 - $P(A | B)$: given that event B happens, the probability that event A happens
- The proof of Bayes' rule is straight-forward (ignoring technical assumptions)

$$P(A \cap B) = P(A | B)P(B) = P(B | A)P(A)$$

Here $P(A \cap B)$ is the probability that events A and ($\cap$) event B both happen

Understanding the World Using Bayes' Rule: Example

Scenario: A student takes a rapid COVID test and it comes back positive.

Assume the following:

- The test accuracy is 95% for infected individuals: $P(\text{Positive} | \text{COVID}) = 0.95$
- The test gives a false positive with 3% chance: $P(\text{Positive} | \text{No COVID}) = 0.03$
- Only 1% of people in the school currently have COVID: $P(\text{COVID}) = 0.01$

But how likely is it that the student is actually infected $P(\text{COVID} | \text{Positive})$?

Solution

Step 1: Bayes' Rule

We want to find $P(\text{COVID} | \text{Positive})$ using

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)},$$

where $A \equiv$ has COVID and $B \equiv$ tests positive

Step 2: Compute total probability of a positive test

$$P(B) = P(B | A) \times P(A) + P(B | \text{not } A) \times P(\text{not } A) = 0.95 \times 0.01 + 0.03 \times 0.99 = 0.0297 = 0.0392$$

Step 3: Apply Bayes' Rule

$$P(A | B) = \frac{0.95 \cdot 0.01}{0.0392} = \frac{0.0095}{0.0392} \approx 0.242$$

Conclusion: Even after a positive test, there's not even 50% chance that the student actually has COVID.

Bayes' Rule in Statistical Modeling

Recall Bayes' rule

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

Let $A \equiv \{\text{True parameter equals } \theta\}$ and $B \equiv \{\text{Observe dataset } \mathcal{D}\}$. We can write

$$P(\theta | \mathcal{D}) = \frac{P(\mathcal{D} | \theta) P(\theta)}{P(\mathcal{D})}$$

- $P(\theta)$: **Prior** distribution about parameter θ
- $P(\mathcal{D} | \theta)$: **Likelihood** of seeing data D if θ were true
- $P(\theta | \mathcal{D})$: **Posterior** distribution *after* seeing data
- $P(\mathcal{D})$: Evidence/normalizing constant

Bayesian vs. Frequentist Rationale

Feature	Frequentist	Bayesian
Parameters (θ)	Fixed but unknown	Random variables with distributions
Data (D)	Random draw from population	Observed fixed evidence
Estimand	Point estimate $\hat{\theta}$	Full posterior $P(\theta D)$
Inference output	Confidence intervals	Full posterior $P(\theta D)$, credible interval

Both are powerful tools for data analysis:

- Use Frequentist methods for convenience and objectivity
- Use Bayesian methods to incorporate prior knowledge, combine multi-view data, handle network data ...

Recipe for Bayesian Analysis

1. Choose a prior $P(\theta)$
 - Encodes what we believe before seeing data
 - Should be chosen before looking at the data to avoid double-counting
2. Write down the likelihood $P(D | \theta)$
 - Comes from the data-generating model (e.g., Binomial for coin flips)
3. Obtain the posterior $P(\theta | D) \propto P(\theta)P(D | \theta)$
 - A. Analytic solution for conjugate pairs of prior and likelihood
 - B. Sampling from the posterior
 - C. Approximation (variational inference)
4. Summarize and infer
 - Posterior mean, median, credible interval
 - Posterior predictive distribution for new data

Examples

1. [One sample test of means \(freq vs. bayes\)](#)
2. [Conjugate priors vs. sampling](#)
3. [Variational inference](#)

Pros & Cons

Pros	Frequentist comparison
Produces intuitive probability statements about parameters	<i>Parameters are not random</i>
Naturally incorporates prior knowledge	<i>Regularization, structured models, ...</i>
Works well with small data sets; uncertainty exists even with $n = 1$	<i>Parametric model still cannot work with $n = 1$</i>
Flexible hierarchical models combine information across groups	<i>Regularization, random effects, ...</i>

Pros & Cons

Cons	Common response
Choosing a good prior may be subjective	Use noninformative priors or priors based on previous studies
Computation can be intensive for complex models	Use variational inference, approximate Bayesian computation, ...

Cons	Common response
Results can be sensitive to strong priors when data are limited	Sensitivity analysis makes the assumption visible
Interpretation may be unfamiliar to audiences used to p-values	Translate posterior statements carefully

Study Report: Prior Sensitivity

- Build a short HTML report showing how Bayesian conclusions change when the prior changes.
- Specify the data-generating model, exact prior distributions, posterior formula, sample sizes, and simulation repetitions.
- Explain what each prior represents and why it is included before interpreting the posterior summaries.
- The report must be readable enough that someone can reproduce the posterior calculation from the description and code.